

variant, and likely future deployment of LTE FDD, also put the world's two largest markets by subscriber numbers onto a different track from mature cell-phone markets."

### Combined functionality

In developing countries like India, where the average revenue per user (ARPU) is far lower than in other countries, firms have to look at new techniques for cost reduction in order to keep a healthy margin for the company's growth. One way to do this is to get more value from their instruments, which is done by getting single units that can perform various kinds of testing.

Shen explains, "India and China also have much lower ARPUs forcing the manufacturers to look for cost savings in R&D. This requires lower-cost protocol analysis tools that may be described as 'one box per engineer' as they combine protocol, functional and RF testing in a single unit to maximise return on test investment. The larger number of combinations of LTE and legacy cellular technologies also increases the amount of testing, further reinforcing the need for flexible cost-effective tools."

"Protocol analysers are largely used to verify protocol compliance. So compliance to various standards is an important feature desired by engineers. Also, protocol analysers that can show the data just as it appears in the specifications will be of great help to engineers working on those protocols. Debugging the protocol layer issues requires a different set of skills. While combining protocol, functional and RF testing might work for some protocols, it may not be efficient while

**Oscilloscope-based protocol analysers have an inherent and unique capability of displaying protocol decoding correlated with the physical-layer analogue waveform. This correlation helps to trace back serial problems to their root cause in the analogue waveform** —Sanchit Bhatia, Digital Applications Specialist, Agilent Technologies

### When is a dedicated protocol analyser needed?

- ▶ **Deep memory.** Applications requiring very deep memory captures require a protocol analyser. Some protocol analysers can have very deep memory of the order of 72 GB, whereas oscilloscopes have maximum 2 GB.
- ▶ **Large number of lanes.** When decoding a large number of data lanes simultaneously in a serial bus, engineers need to use a protocol analyser. They can see all the 16 lanes in a PCIe x16 link by using a protocol analyser, whereas oscilloscopes decode one lane at a time.
- ▶ **Specialised capabilities.** Protocol analysers often have specialised performance analysis capabilities that go well beyond just triggering and decoding. For example, in case of PCI Express, one would need a protocol analyser to analyse a state machine called LTSSM.
- ▶ **Jammers and exercisers.** Some applications might require serial link jamming, which means deliberate insertion of errors in the link. In this case, a jammer would be needed. Also, some applications might require handshakes between the instrument and the device under test, in which case an exerciser might be needed. Jammers and exercisers are normally used with dedicated protocol analysers, not with oscilloscopes.

—Sanchit Bhatia, Digital Applications Specialist, Agilent Technologies

debugging serial data solutions like PCIe and SAS," shares Prabhajana Rao, regional sales manager-India and SE Asia, ANZ, Teledyne LeCroy Corporation.

For the last few years, there has been an increase in the use of oscilloscopes for protocol decoding of serial buses as well. Initially, these were used for protocol decoding of low-speed buses like I<sup>2</sup>C and SPI, but the protocol applications of scopes have now expanded to incorporate many more buses like CAN, LIN, MIL-STD-1553, SATA, USB 2.0,

USB 3.0, and all generations of PCI Express. Since oscilloscopes are already used for physical-layer debug of serial buses, the protocol analysis capability adds value by having a single instrument that provides the capabilities of an oscilloscope and protocol analyser.

Sanchit Bhatia, digital applications specialist, Agilent Technologies, explains the various advantages of using oscilloscopes for protocol analysis: "First, this saves the bench space and investment in a separate protocol analyser for each serial bus. Second, using a single instrument saves time and effort associated with the learning curve for each protocol analyser is removed. Third, oscilloscope-based protocol analysers have an inherent and unique capability of displaying protocol decoding correlated with the physical-layer analogue waveform. This correlation helps to trace back serial problems

**User interfaces for these tools are becoming more and more user-friendly in the way they display decoded data, and hence protocol intricacies can now be dealt easily. It makes the task of test engineers fast and easy while maintaining the accuracy of testing** —Siddharth Patel, Lead Engineer for Embedded Software, EATON India Engineering Centre